

Table 1 Proteases employed for production of fish muscle protein hydrolysates

Fish species	Protease	Protease type	Cleavage site of protease	Hydrolysis conditions	Peptides/ AAs in fraction	Reference
Monkfish	Trypsin	Serine endopeptidase	Lys or Arg residues	40°C, pH 8.0, E/S 2% with 4 h	EWPAQ, FLHRP, LMGQW	Chi et al. (2014)
Salmon	Pepsin Corolase PP	Endopeptidase Endopeptidase (metalloprotease and serine protease)	Between hydrophobic AA residues Broad substrate specificity	15 U/g protein, 37°C, 120 min 31.2 U/g protein, 37°C, 60 min	IVY, VW, IY, IW, VY, TVY, VFPS, VTVNPYKWLP, IWHHT, YALPHA, ALPHA	Darewicz et al. (2014)
Cod (<i>Gadus morhua</i>)	Flavourzyme® Alcalase®	Mixture of endo- and exoproteases Serine endopeptidase	Broad substrate specificity Broad substrate specificity	50°C, 0.1% Flavourzyme® or Alcalase®	Total AA content (52-92% of the total crude proteins) Free AA (28- 62% of the total AAs)	Dauksas et al. (2004)
Cobia (<i>Rachycentron canadum</i>)	Alcalase® Flavorase Protamex®	Serine endopeptidase Mixture of endo- and exoproteases Serine endopeptidase	Hydrophobic AA Broad substrate specificity Broad substrate specificity	Alcalase®(pH 8.0, 50°C, 99.75 U/g) Flavourzyme® (pH 7.0, 50°C, 2.07 U/g) Protamex® (pH 7.0, 40°C, 8.41 U/g)	-	Fonseca et al.(2016)
Catfish (<i>Pangasius hypophthalmus</i>)	Papain Bromelain Neutrase®	Cysteine endopeptidase Cysteine endopeptidase Metalloprotease	After Arg or Lys preceded by a hydrophobic AA not followed by Val After Lys, Ala or Tyr Hydrophobic AA	Papain (55°C, pH 7.5, 180 min, 36.2 g/L), Bromelain (55°C, pH 6.5, 180 min, 135.8 g/L) Neutrase® (50°C, pH 6.5, 180 min, 108.4 g/L)	Papain: Arg (12.64 ± 0.06 g/kg protein) Bromelain: Lys (22.46 ± 0.10 g/kg protein) Neutrase®: Lys (18.86 ± 0.07 g/kg protein)	Ha et al. (2017)
Rainbow trout	Pepsin Trypsin α-chymotrypsin Alcalase®	Endopeptidase Serine endopeptidase Serine endopeptidase Serine endopeptidase	Between hydrophobic AA residues Lys or Arg residues Medium to large sized hydrophobic AA (Typ, Phe, Trp) Broad substrate specificity	Pepsin (37°C, pH 2.0), Trypsin (37°C, pH 7.0), α- chymotrypsin (37°C, pH 7.0), Alcalase (50°C, pH 7.0), Neutrase (50°C, pH	Peptide purified from peptic fraction: LVNGPAMSPNAN	Kim and Byun (2012)

	Neutrase® Papain	Metalloprotease Cysteine endopeptidase	Hydrophobic AA. After Arg or Lys preceded by a hydrophobic AA not followed by Val	7.0), papain (37°C, pH 7.0), E/S 1/100 (w/w) for 6 h		
Under-utilised skipjack tuna (<i>Katsuwonus pelamis</i>)	Alcalase® Flavourase Neutrase® Trypsin Protamex®	Serine endopeptidase Metalloprotease Metalloprotease Serine endopeptidase Serine endopeptidase	Broad substrate specificity - Hydrophobic AA Lys or Arg residues Hydrophobic AA	Alcalase® (pH 8.0, 55°C), Flavourase (pH 7.0, 55°C), Neutrase® (pH 7.0, 50°C), Trypsin (pH 8.0, 50°C), Protamex® (pH 7.0, 50°C), 20,000 U/g, 5h	All fractions rich in His (489.14 - 1337.90 mg/100g) and Pro (144.36 - 170.50 mg/100g) Trypsin and Protamex®: 80% soluble protein at 2.5h	Liu et al. (2015)
Bighead croaker (<i>Collichthys niveatus</i>)	Flavourzyme® Alcalase® Neutrase® Protamex®	Mixture of endo- and exoproteases Serine endopeptidase Metalloprotease Serine endopeptidase	Broad substrate specificity Broad substrate specificity Hydrophobic AA Hydrophobic AA	Protease activity 10.103 U, 4h hydrolysis, pH and temperature not reported.	Neutrase®: SUA (88.259 µg/ mL) Alcalase®: SUA (65.129 µg/ mL) Protamex®: SUA (40.796 µg/ mL) Flavourzyme®: SUA (49.172 µg/ mL)	Shen et al. (2012)
Catfish (<i>C.batrachus</i>)	Alcalase® Papain	Serine endopeptidase Cysteine endopeptidase	Broad substrate specificity After Arg or Lys preceded by a hydrophobic AA not followed by Val	Alcalase® (pH 8, 55°C), Papain (pH 7, 55°C). E/S ratio 1:100 (w/w), 30-180 min.	Alcalase®: Glu (15.66 ± 0.22%), Lys (10.35 ± 0.09%), Asp (9.86 ± 0.01%) Papain: Glu (16.40 ± 0.22%), Lys (10.96 ± 0.39%), Asp (10.21 ± 0.25%)	Seniman et al. (2014)

AA, amino acid; E/S, enzyme/substrate; SUA, sweet and umami taste amino acids (Ala, Asp and Glu)